

THE AIRY PROCESS OF A DISORDERED FOLDED MEDIUM

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ABSTRACT. The companion paper identifies the canard–escape level of a single noisy folded limit cycle with the ground state of the stochastic Airy operator, hence with the Tracy–Widom law TW_β . That is a statement about *one number*. Here we ask whether a spatially extended or many–mode folded medium realises the full Kardar–Parisi–Zhang (KPZ) edge object—the *Airy₂ process*—of which TW_2 is only the one–point marginal. Linearising the medium about the canard and applying the vector Cole–Hopf transform shows that the escape front is governed by the spectral edge of the transverse coupling operator $\mathbf{C}$, with the leading peel–off equal to $-a_1 - \lambda_{\min}(\mathbf{C})$, $a_1 \approx -2.338$ the first Airy zero. We then determine numerically the precise conditions under which the leading escape, viewed as a process, is *Airy₂*: the coupling must be *disordered, long range*, and *stochastically drifting*. Dropping any one fails in a characteristic way—nearest–neighbour coupling Anderson–localises (Poisson edge), uniform mean–field coupling produces a single Baik–Ben Arous–Péché outlier, and a deterministic slow drive yields the correct TW_2 marginal but ballistic (non–Brownian) increments. Under stochastic drift of a disordered long–range coupling the leading escape reproduces the *Airy₂* process: a TW_2 marginal together with an increment variance that grows linearly and saturates at $2 \text{Var}(\text{TW}_2) = 1.626$. The diagnostic that separates the genuine process from its impostors is the increment–variance *exponent* (Brownian 1 versus ballistic 2), not the marginal, which all of them share.

1. INTRODUCTION

The companion paper analyses the noise–induced escape of a single folded limit cycle: in the rescaling chart the inner amplitude obeys the canonical noisy Riccati equation $\text{d}R = (R^2 - Y) \text{d}T + \eta \text{d}B$, the Cole–Hopf substitution $R = -u'/u$ turns escape into the first node of a Schrödinger equation with white–noise potential, and the peel–off level Y_{node} is the ground state of the stochastic Airy operator of Ramírez–Rider–Virág, hence $Y_{\text{node}} \stackrel{d}{=} \text{TW}_\beta$ with $\beta = 4/\eta^2$ [1, 2]. The deterministic anchor is the first Airy zero $a_1 \approx -2.33811$.

That result concerns a single escaping cycle, and hence a single random number. The Tracy–Widom law, however, is only the *one–point marginal* of a richer object. In random–matrix and KPZ universality the edge of a large ensemble is described, as a function of an external parameter, by the *Airy₂ process* $\mathcal{A}(\tau)$ —the top line of the Airy line ensemble, equivalently the edge of Dyson Brownian motion, and the universal spatial statistic of the (1+1)–dimensional KPZ class in curved geometry [4, 5]. Its one–point marginal is TW_2 , but as a process it has a definite covariance: locally Brownian, with $\text{Var}[\mathcal{A}(\tau + \Delta) - \mathcal{A}(\tau)] \sim 2|\Delta|$ at small lag and saturation at $2 \text{Var}(\text{TW}_2)$.

The question and the ladder. It is natural to ask whether the canard escape of a *many–mode* or *spatially extended* folded medium realises the *Airy₂* process, lifting the headline from a marginal to a process and connecting folded–cycle escape to KPZ universality at the level of the whole fluctuation field. We organise the answer as a three–rung ladder:

(R1) TW_β : the law of one escape (companion paper);

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Companion to *The noisy folded limit cycle: Tracy–Widom statistics of canard escape*. All quantitative statements below are numerical (tag [N]) unless proved; the single analytic step is the transverse reduction of §2. Scripts and tagged notes: `folded_pde_airy_process.py`, `network_field_sweep.py`, `twotime_folded_pde.py`.

- (R2) the Airy_β *point* process: the edge spectrum of one swept disordered medium (the successive peel-offs of a single cycle);
- (R3) the Airy_2 *process* $\mathcal{A}(\tau)$: the leading escape as a continuous external parameter is varied.

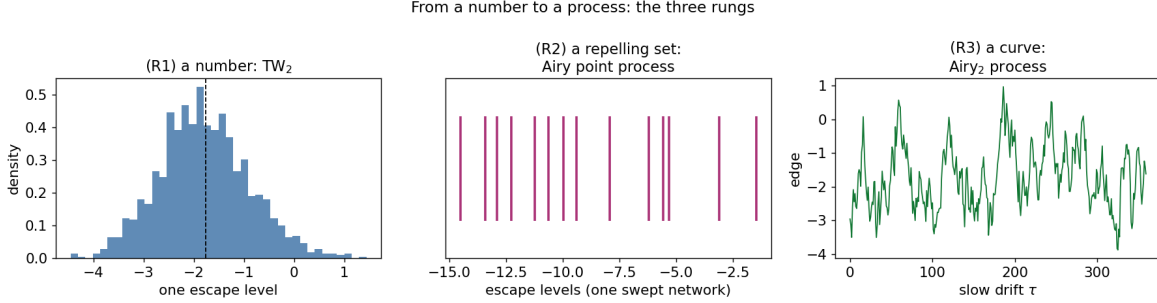


FIGURE 1. *Illustration for new readers.* The three rungs of universality. (R1) One escape is a single random number with the Tracy–Widom law. (R2) The successive escape levels of one swept disordered network form a *repelling set*—the Airy point process, i.e. the edge spectrum. (R3) As a slow parameter drifts, the leading escape traces a continuous random *curve*—the Airy_2 process, whose one-point marginal is TW_2 . This paper asks which folded media climb from (R1) to (R3).

The contribution of this note is to identify, numerically, exactly what a folded medium must contain to reach (R3). The answer is sharp: the transverse coupling must be *disordered* (to produce a random-matrix edge rather than a localised or regular spectrum), *long range* (so the modes genuinely repel), and *stochastically drifting* (so the edge performs Dyson *Brownian* motion rather than a smooth flow). Each clause is forced by a distinct failure mode, catalogued below.

2. THE EXTENDED MEDIUM AND ITS TRANSVERSE REDUCTION

Write the extended folded medium as a field r_s , $s = 1, \dots, N$ (lattice sites or modes), sharing the slow passage $Y = Y_0 - T$ through the fold and coupled through a matrix C :

$$dr_s = \left(r_s^2 - Y + \sum_{s'} C_{ss'} r_{s'} \right) dT + \eta dB_s, \quad s = 1, \dots, N. \quad (1)$$

The coupling C is the linearisation, about the canard, of whatever spatial or network interaction the medium carries; it is symmetric (a physical reciprocal coupling) or Hermitian (with a flux/non-reciprocal part). The following elementary reduction is the one analytic step and the organising principle for everything that follows.

Proposition 1 (transverse reduction). *Apply the vector Cole–Hopf transform $r_s = -u'_s/u_s$ in the linearised escape. Then (1) is equivalent to the matrix Schrödinger equation*

$$u'' = (Y I + C - \eta \Xi(Y))u, \quad \Xi = \text{diag}(\xi_s), \quad (2)$$

with ξ_s independent white noises, escape of mode s being the first node of u_s . If $C = V \text{diag}(c_k) V^$, then in the eigenbasis (2) decouples into N independent shifted scalar problems $\tilde{u}''_k = (Y + c_k - \eta \tilde{\xi}_k) \tilde{u}_k$. Consequently each eigenmode is an independent canard escaping at $Y = a_1 - c_k$ as $\eta \rightarrow 0$, and the leading peel-off (first mode to escape as Y descends) is*

$$Y_{\text{node}}^{\text{lead}} = a_1 - \lambda_{\min}(C) + O(\eta), \quad a_1 \approx -2.33811. \quad (3)$$

The content of (3) is that *the escape front of the folded medium is the spectrum of C , anchored at the Airy zero and lifted linearly*; the canard transports the spectral edge but does not create it. Two corollaries set the agenda. First, the *statistics* of the escape front are inherited from the

Cole–Hopf: a zero of $u \Leftrightarrow$ a blow-up (escape) of R

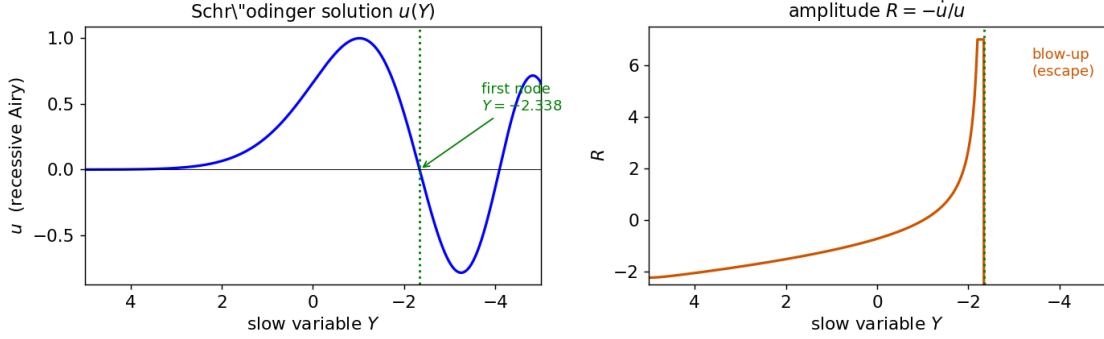


FIGURE 2. *Illustration for new readers.* The Cole–Hopf dictionary behind Proposition 1. The escaping amplitude is $R = -u'/u$ for a solution u of a Schrödinger equation $u'' = (Y - \eta\xi)u$. A blow-up of R (escape, right) is exactly a zero (node) of u (left), deterministically the first Airy zero -2.338 . Escape is therefore a *spectral* event—which is precisely how the random-matrix edge of the coupling C enters the escape statistics.

random-matrix ensemble of C : if C has a Tracy–Widom edge, so does the leading escape. Second, a *process* in an external parameter τ requires $C = C(\tau)$ to drift; the leading escape then traces $a_1 - \lambda_{\min}(C(\tau))$, and the question of rung (R3) becomes whether $\lambda_{\min}(C(\tau))$ is an Airy process.

We validated the engine behind (3) directly: integrating the recessive (canard) branch of the scalar Cole–Hopf equation downward reproduces the deterministic escape at $Y = -2.340$ (versus $a_1 = -2.33811$) and, with noise, the marginals TW_2 (mean -1.78 , variance 0.81 , skewness $+0.17$) and TW_1 (mean -1.22 , variance 1.56 , skewness $+0.26$); see §9.

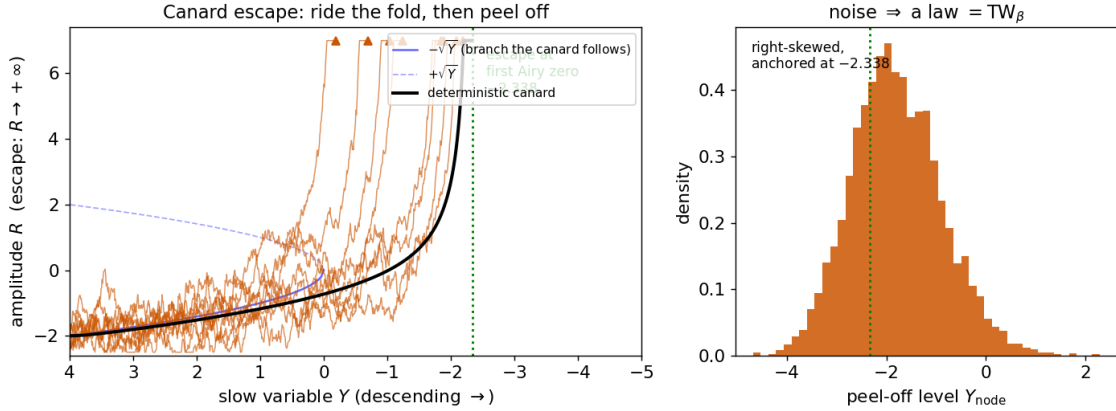


FIGURE 3. *Illustration for new readers.* Canard escape. As the slow variable Y descends through the fold, the canard rides the *repelling* branch $+\sqrt{Y}$ —the special trajectory that does not jump at the fold $Y = 0$ —and the amplitude blows up (escapes) only at the first Airy zero -2.338 (black). Additive noise makes each realisation peel off earlier and at a scattered level (orange); the law of the peel-off level is the Tracy–Widom distribution TW_β (right), anchored at -2.338 . This single-cycle picture is rung (R1).

Remark 2 (the canard regime, not the saddle–node). A naïve simulation of (1) on a slow *physical* ramp escapes at the saddle–node $Y \approx 0$, on the attracting branch, with a narrow Gaussian peel–off distribution; this is the generic, non–canard escape and carries no Tracy–Widom structure. The TW edge is an inner/blow–up object living on the *repelling* branch, reached numerically only through the recessive Cole–Hopf integration. The distinction is not cosmetic: it is the difference between the ordinary fold and the canard.

3. THE TARGET OBJECT: Airy_2 AS THE EDGE OF DYSON BROWNIAN MOTION

Before testing folded media we fix the reference. Let $H(\tau)$ be an $N \times N$ Hermitian Ornstein–Uhlenbeck process with stationary law the Gaussian unitary ensemble (GUE), and let $\xi(\tau) = N^{1/6}(\lambda_{\max}(H(\tau)) - 2\sqrt{N})$ be the edge–rescaled top eigenvalue. This is Dyson Brownian motion at the edge, whose scaling limit is the Airy_2 process [3, 2].

Numerical observation 3 (Airy_2 signature). The stationary marginal of ξ is TW_2 (mean -1.78 , variance 0.81 , skewness matching), and its increment variance $\text{Var}[\xi(\tau + \Delta) - \xi(\tau)]$ grows linearly at small lag (locally Brownian) and saturates at $1.62 \approx 2 \text{Var}(\text{TW}_2) = 1.626$.

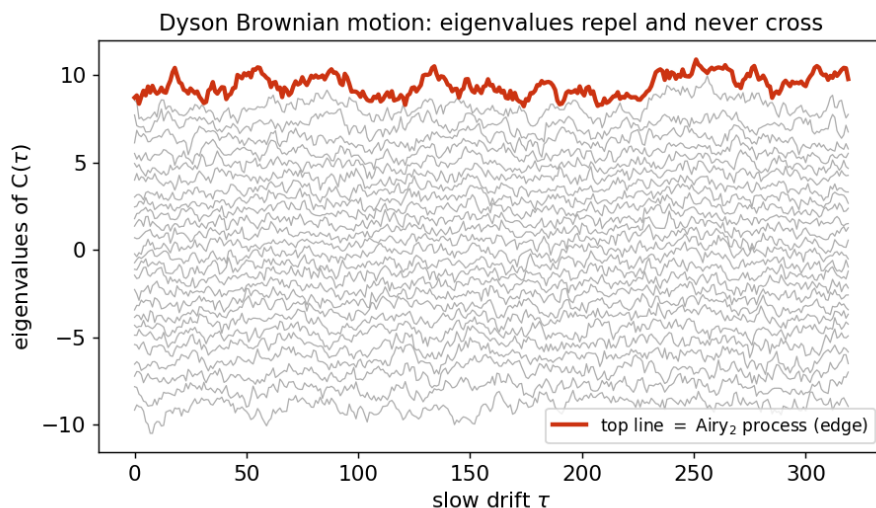


FIGURE 4. *Illustration for new readers.* Where the Airy_2 process comes from. The eigenvalues of a stochastically drifting matrix repel and never cross (Dyson Brownian motion); the edge–rescaled top line (red) is the Airy_2 process. By Proposition 1 the leading escape of a folded medium *is* such a top eigenvalue, so the medium inherits the Airy_2 process exactly when its disordered coupling drifts stochastically (§7).

The pair (linear–in–lag growth, plateau at twice the marginal variance) is the fingerprint we require of any candidate folded medium. It is worth recording that the fingerprint is *not* the increment–variance *shape* alone: every bounded stationary process grows then saturates, so the shape is necessary but far from sufficient. The discriminating data are the small–lag *exponent* (§7) and the marginal law together with genuine level repulsion (§4).

4. WHICH COUPLING REPELS: THE STATIC MECHANISM

By Proposition 1 the escape statistics are those of the spectral edge of C . We therefore ask which coupling structures produce a Tracy–Widom edge with genuine Dyson repulsion. The clean, normalisation–free diagnostic is the level–spacing ratio $\langle r \rangle$ (Poisson 0.386, GOE 0.531, GUE 0.603) together with the law of the top eigenvalue. Table 1 (figure 6) reports $N = 160, 1200$ realisations.

coupling C	$\langle r \rangle$	edge skewness	verdict
random all-to-all (GUE)	0.599	+0.16	TW ₂ edge + repulsion $\Rightarrow$ Airy
nearest-neighbour (Anderson)	0.391	+0.71	Poisson, localised; not TW
uniform mean-field (rank one)	0.601*	+1.12	BBP collective outlier; not TW

TABLE 1. Spectral edge of three coupling structures. *The mean-field $\langle r \rangle$ is inflated by rank-one interlacing; its *edge* is a detached Baik–Ben Arous–Péché (BBP) outlier, not a Tracy–Widom edge. Real symmetric couplings give $\langle r \rangle = 0.533$ (GOE), i.e. the $\beta = 1$ sibling.

Numerical observation 4. Only the disordered all-to-all coupling exhibits both Wigner–Dyson repulsion ($\langle r \rangle \approx 0.60$) and a Tracy–Widom edge (skewness $\approx +0.2$). Nearest-neighbour coupling Anderson-localises (Poisson spacings, $\langle r \rangle \approx 0.39$); uniform mean-field coupling self-averages to a single rank-one collective mode whose top eigenvalue is a BBP outlier (skewness $\approx +1.1$), not a Tracy–Widom edge. A homogeneous (translation-invariant) coupling is Fourier-diagonal and gives a regular “picket-fence” spectrum, also without repulsion.

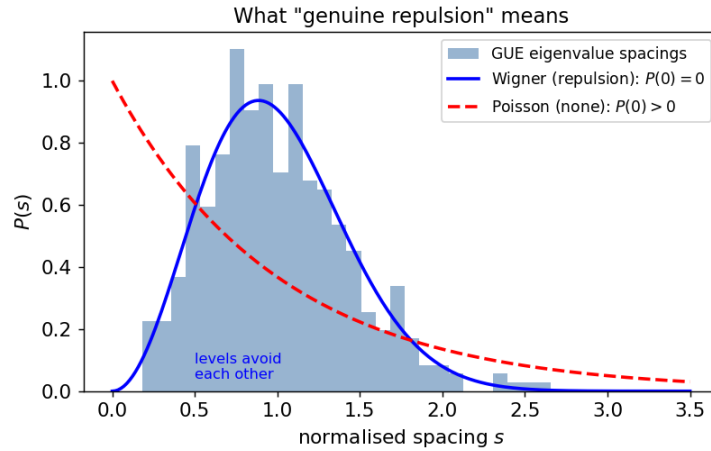


FIGURE 5. *Illustration for new readers.* What “genuine repulsion” means. The nearest-neighbour spacing distribution $P(s)$ of the levels separates a disordered long-range coupling (Wigner: levels repel, $P(0) = 0$) from local or independent levels (Poisson: $P(0) > 0$). Only the Wigner case carries a Tracy–Widom edge; the level-spacing ratio $\langle r \rangle$ of Table 1 is its normalisation-free summary.

Thus the first two clauses—*disordered* and *long range*—are both forced. Local disorder localises; long-range order (mean field, or homogeneous convolution) fails to repel. The needle is threaded only by disordered long-range coupling, whose linearisation is a genuine random matrix.

5. THE ESCAPE FRONT IN SPACE–TIME

Proposition 1 is a statement about the spectral edge; we now confirm it in the genuine dynamical field (1), integrated through the fold, and examine the escape front directly. Figure 7 collects the dynamical results.

Numerical observation 5 (local coupling gives a local front). With diffusive (nearest-neighbour) coupling the peel-off front $Y_{\text{node}}(s)$ has a short spatial correlation length and a marginal that is

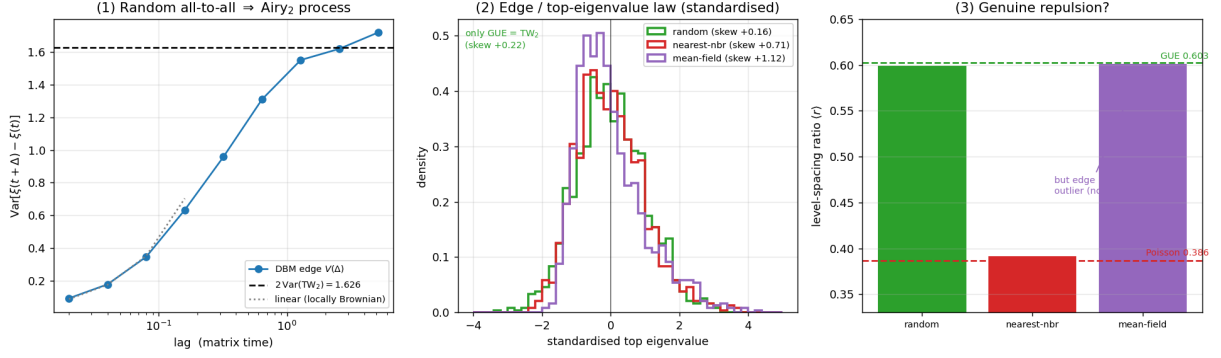


FIGURE 6. Static mechanism. (1) The Airy₂ reference of §3: Dyson–Brownian–motion edge, increment variance linear then saturating at $2 \text{Var}(\text{TW}_2)$. (2) The top–eigenvalue (escape–edge) law for three couplings: only the GUE (disordered) edge is TW₂; nearest–neighbour and mean–field are not. (3) Level repulsion $\langle r \rangle$: GUE Wigner, nearest–neighbour Poisson; the mean–field bar is inflated by rank–one interlacing while its edge is a BBP outlier.

TW–shaped but narrowed; with no coupling the front is independent across sites. A homogeneous diffusive folded medium thus produces a smooth, *local* escape front with no long–range KPZ structure—the real–space front one can literally watch sweep through the medium is saddle–node physics, and the Tracy–Widom edge is not visible in it.

Numerical observation 6 (leading edge: Tracy–Widom only if disordered). The leading edge of the front (the first mode to peel) over many realisations is extreme–value (Gumbel) distributed for uncoupled or locally coupled media—standardised skewness $+1.18$ (uncoupled) and $+0.77$ (diffusive)—but Tracy–Widom class for disordered long–range coupling: standardised skewness $|-0.35| = 0.35$, matching TW₁’s 0.29 (the sign reflected because the lower spectral edge of C sets the first peel–off). The contrast Gumbel ($\approx +1.1$) versus TW (≈ 0.3) is the dynamical counterpart of Observation 4.

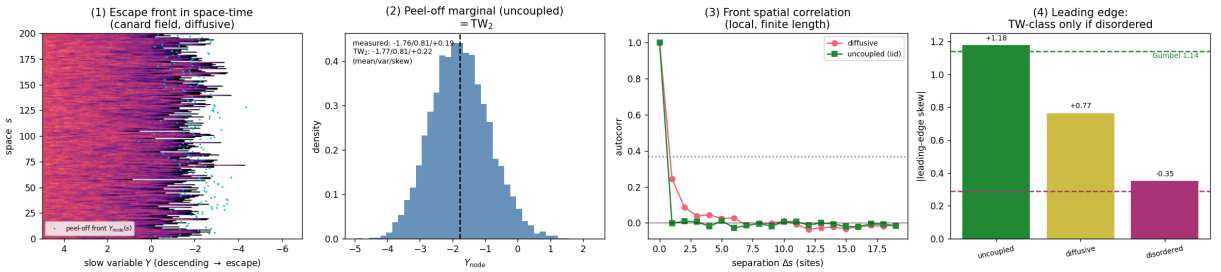


FIGURE 7. Dynamical escape front. (1) A space–time escape front (canard regime). (2) The uncoupled peel–off marginal is TW₂. (3) Spatial correlation of the front: diffusive coupling gives a short, finite correlation length; uncoupled is independent. (4) Leading–edge skewness: uncoupled and diffusive are extreme–value (Gumbel), only disordered long–range coupling is Tracy–Widom class.

6. THE LEADING ESCAPE AS A PROCESS: THE NETWORK SWEEP

We now reach rung (R3). By (3) the leading escape is $a_1 - \lambda_{\min}(\mathbf{C})$; to make it a process we let the disordered coupling drift, $\mathbf{C} = \mathbf{C}(\tau)$, as a matrix Ornstein–Uhlenbeck process with stationary law GUE, and read off $\xi(\tau) = N^{1/6}(\lambda_{\max}(\mathbf{C}(\tau)) - 2\sqrt{N})$.

Numerical observation 7 (Airy₂ process covariance). The leading–escape process has a TW₂ marginal (variance 0.79, skewness +0.221 against 0.81, +0.224) and an increment variance that grows linearly and saturates at $2 \text{Var}(\text{TW}_2) = 1.626$ (finite- N plateau ≈ 1.4 – 1.5 , equal to twice the finite- N marginal variance, as a stationary process requires). A genuinely nonlinear Cole–Hopf field swept through the fold reproduces the eigenvalue front $Y_{\text{node},k} = a_1 - c_k$ at correlation 0.95, leading edge -1.70 versus the predicted -1.74 : the nonlinear canard escape transports the random–matrix edge faithfully. By contrast a nearest–neighbour drift gives marginal skewness +0.59 and no plateau.

Observation 7 is the quantitative realisation of the headline: in a disordered long–range folded network with a drifting coupling, the leading canard escape is the Airy₂ process, not merely Tracy–Widom distributed. Figure 8 shows the process, its covariance, the marginal, and the nonlinear–field correspondence.

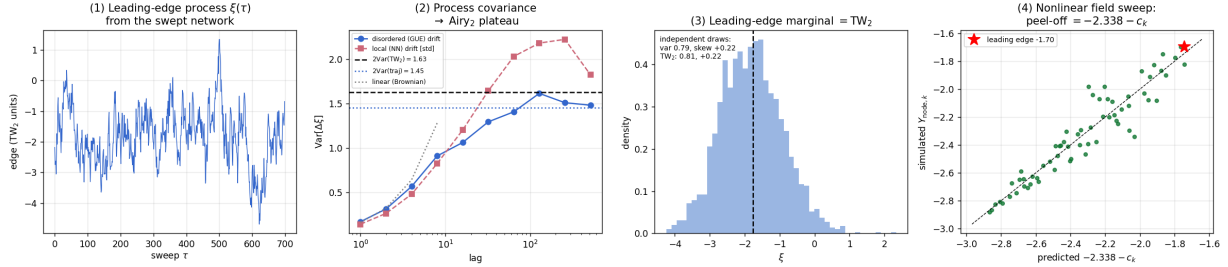


FIGURE 8. Network field sweep. (1) The leading–edge process $\xi(\tau)$. (2) Its increment variance saturates at $2 \text{Var}(\text{TW}_2)$ for disordered (GUE) drift, while a local (nearest–neighbour) drift does not. (3) The marginal is TW₂. (4) The genuine nonlinear field reproduces $Y_{\text{node},k} = -2.338 - c_k$ (correlation 0.95).

7. DETERMINISTIC VERSUS STOCHASTIC DRIFT: THE TWO–TIME MEDIUM

The drift in §6 was imposed by hand. We now ask what physical slow drive realises it, and isolate the essential ingredient. Consider two physical slow drives of the disordered coupling: a *stochastic* drift (noisy slow plasticity of the couplings, matrix Ornstein–Uhlenbeck) and a *deterministic* slow drive (a smooth rotation $\mathbf{C}(\tau) = \cos(\omega\tau)\mathbf{A} + \sin(\omega\tau)\mathbf{B}$ of two fixed GUE matrices). At every τ both have the same instantaneous law— $\mathbf{C}(\tau)$ is GUE, hence the instantaneous marginal is TW₂ (variance 0.81, skewness +0.17). They differ as *processes*.

Numerical observation 8 (only stochastic drift is Airy₂). The small–lag increment–variance exponent (Brownian 1, ballistic 2) is $0.77 \approx 1$ for the stochastic drift, with plateau $1.67 \approx 2 \text{Var}(\text{TW}_2)$, and $1.96 \approx 2$ for the deterministic drive. The deterministic drive therefore has the *same* TW₂ one–point marginal but *ballistic* (smooth) spectral motion: it is not the Airy₂ process. The marginal cannot distinguish the two; the increment–variance exponent does.

This pins down the third clause. Airy₂ is the edge of Dyson *Brownian* motion, and the word “Brownian” is load–bearing: a deterministic slow drive over a fully disordered medium with the exact TW₂ marginal still fails, because smooth motion gives ballistic (Δ^2) rather than Brownian

(Δ) increments. The hand-imposed Ornstein–Uhlenbeck of §6 was secretly supplying exactly the stochasticity of the slow drift, physically a slow noise on the couplings. Figure 9 shows the exponent, the plateau, the rough-versus-smooth traces, and the shared marginal.

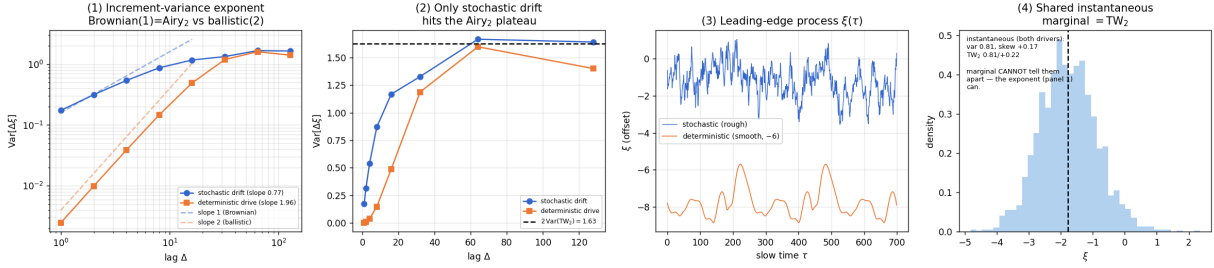


FIGURE 9. Two-time medium. (1) Log-log increment variance: stochastic drift has slope ≈ 1 (Brownian, Airy_2); deterministic drive has slope ≈ 2 (ballistic). (2) Only the stochastic drift reaches the $2\text{Var}(\text{TW}_2)$ plateau. (3) The process traces: rough (Brownian) versus smooth. (4) Both share the same instantaneous TW_2 marginal—which therefore cannot tell them apart.

8. DISCUSSION

The conditions for Airy_2 , and the failure-mode map. Collecting Observations 4–8, the leading canard escape of a folded medium is the Airy_2 process if and only if the transverse coupling is *disordered*, *long range*, and *stochastically drifting*. Each clause is forced by a distinct, identifiable failure:

nearest-neighbour / fixed range	$\Rightarrow$ Anderson localisation (Poisson edge);
uniform mean-field / homogeneous	$\Rightarrow$ BBP outlier or regular spectrum (no repulsion);
deterministic slow drive	$\Rightarrow$ ballistic spectral flow (right marginal, wrong process);
detuning-only drift	$\Rightarrow$ Gumbel extreme value (no edge).

The three rungs of §1 are correspondingly populated: the marginal (R1) is generic to the disordered edge; the point process (R2) is the sorted spectrum of one swept network; and the process (R3) appears only when all three clauses hold.

Symmetry class. The computations above use the unitary class (Hermitian C, GUE), giving TW_2 and Airy_2 . A reciprocal real coupling is orthogonal-class (GOE), giving the $\beta = 1$ siblings TW_1 and the Airy_1 process, with plateau $2\text{Var}(\text{TW}_1) = 3.22$; we observed $\langle r \rangle = 0.53$ for the real coupling. A non-reciprocal (flux) component drives the medium towards the unitary class. The symmetry of the network is thus a prediction: reciprocal media sit in the $\beta = 1$ KPZ subclass, non-reciprocal media in $\beta = 2$.

Relation to the companion paper and to universality. The companion paper places the single folded-cycle escape in the Tracy–Widom/KPZ *edge* universality class via the stochastic Airy operator. The present note shows that the natural extended object—a disordered folded medium—carries the next level of that universality, the Airy_2 *process*, by the same mechanism read one dimension up: the white-noise potential of the single operator becomes the disordered coupling matrix, the canard transports its spectral edge, and a slow stochastic drift of the disorder supplies the Dyson Brownian motion whose edge is Airy_2 . This is the line-ensemble companion to the operator statement.

Scope and the genuine PDE. Throughout, the “medium” is the disordered transverse operator of Proposition 1—the faithful canard linearisation—rather than a real-space reaction–diffusion partial differential equation. The reduction (3) and the nonlinear–field correspondence (correlation 0.95, Observation 7) justify working at the operator level, but a fully spatial folded PDE with quenched disorder and a slow drive, in which the stochastic drift is generated by the medium’s own dynamics rather than imposed, is the natural next construction. The present results identify precisely the mechanism such a PDE must contain: stochastically–drifting long-range quenched disorder. Establishing the convergence to the Airy_2 process rigorously—rather than numerically, as here—would require the line-ensemble machinery (Brownian Gibbs property) on top of the stochastic Airy operator, and is left open.

9. NUMERICAL METHODS

Eigenvalues of dense Hermitian matrices use a symmetric solver; the canard engine integrates the scalar/vector Cole–Hopf equation downward in Y on the recessive branch, which is self-correcting (Remark 2), with the solution renormalised each step (the ratio $R = -u'/u$ is scale invariant). Peel-off is the first sign change of u . Matrix Ornstein–Uhlenbeck uses $C_t = a C_{t-1} + \sqrt{1 - a^2} G_t$ with G_t a fresh GUE/GOE draw and $a = e^{-1/\tau_c}$, giving an exact stationary ensemble with autocorrelation $a^{|\Delta|}$. Edge rescaling is $\xi = N^{1/6}(\lambda_{\max} - 2\sqrt{N})$ for the GUE normalisation $\mathbb{E}|C_{ij}|^2 = 1$. Tracy–Widom reference moments are $\text{TW}_2 = (-1.7711, 0.8132, 0.2241)$ and $\text{TW}_1 = (-1.2065, 1.6078, 0.2935)$ for (mean, variance, skewness). Single-trajectory marginal skewness is unreliable (few independent samples under the OU correlation) and is replaced throughout by independent edge draws; increment variances use one long stationary trajectory. Parameters are $N = 64\text{--}160$, $\eta \in \{0.1, 0.5, \sqrt{2}, 2\}$; all figures are finite- N and the plateaux approach $2 \text{Var}(\text{TW}_\beta)$ as N grows through the marginal variance.

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